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United States Patent	12404973
Kind Code	B2
Date of Patent	September 02, 2025
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Tank

Abstract

A tank includes a liner and a reinforcement layer. The liner has a body portion with a cylindrical shape, and a pair of dome-shaped portions respectively provided at both ends of the body portion in an axial direction of the body portion. The reinforcement layer covers the liner and is made of a fiber reinforced resin that is a resin-impregnated fiber bundle. The reinforcement layer includes a part reinforcement layer disposed from an apex side of each of the dome-shaped portions across a boundary between the dome-shaped portion and the body portion to part of the body portion, and an outer hoop layer disposed outside the part reinforcement layer and provided so as to press an end, adjacent to the body portion, of the part reinforcement layer.

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Appl. No.: **18/455586**

Filed: **August 24, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240093834 A1	Mar. 21, 2024

Foreign Application Priority Data

JP	2022-147978	Sep. 16, 2022

Publication Classification

Int. Cl.: **F17C1/06** (20060101)

U.S. Cl.:

CPC **F17C1/06** (20130101); F17C2201/0109 (20130101); F17C2203/012 (20130101); F17C2203/0604 (20130101); F17C2203/0624 (20130101); F17C2203/067 (20130101); F17C2203/0673 (20130101); F17C2221/012 (20130101); F17C2221/033 (20130101); F17C2221/035 (20130101); F17C2260/011 (20130101); F17C2270/0168 (20130101); F17C2270/0184 (20130101)

Field of Classification Search

CPC: F17C (1/06); F17C (2201/0109); F17C (2203/012); F17C (2203/0604); F17C (2203/0624); F17C (2203/067); F17C (2203/0673); F17C (2260/011); F17C (2270/0168)

USPC: 220/590; 220/581; 220/588; 220/589

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. 2022-147978 filed on Sep. 16, 2022, incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(2) The disclosure relates to a tank.

2. Description of Related Art

(3) A tank, such as a hydrogen tank, is described in, for example, Japanese Unexamined Patent Application Publication (Translation of PCT Application) No. 2019-527321 (JP 2019-527321 A). The tank described in JP 2019-527321 A includes a liner and a reinforcement layer. The liner has a body portion and a pair of dome-shaped portions. The body portion has a cylindrical shape. The dome-shaped portions are respectively provided at both ends of the body portion in an axial direction of the body portion. The reinforcement layer is provided so as to cover the liner. The reinforcement layer is formed by winding a resin-impregnated fiber bundle around the outer periphery of the liner by a filament winding (FW) method. More specifically, the reinforcement layer is made up of a first tape and a second tape. The first tape is placed so as to be wound around the liner by helical winding or hoop winding. The second tape is placed within a range shorter than the diameter of the body portion so as to pass around at least the dome-shaped portions.

SUMMARY

(4) In the tank, the strength of each dome-shaped portion (in other words, the strength of the tank in the axial direction) is ensured by helical winding, and the strength of the body portion (in other words, the radial strength of the tank) is ensured by hoop winding. Helical winding is a method of spirally winding a fiber bundle around the outer peripheries of the dome-shaped portions and the body portion, so the fiber bundle formed by helical winding is formed not only on the dome-shaped portions but also on the body portion. However, the fiber bundle formed by helical winding contributes to reinforcement of the body portion just a little.

(5) In the tank described in JP 2019-527321 A, a fiber bundle is continuously wound so as to shuttle between the dome-shaped portions respectively provided at both ends of the liner by helical winding, so the fiber bundle definitely passes across the body portion. Therefore, there is an inconvenience that some usage of fiber reinforced resin, that is, a resin-impregnated fiber bundle, is wasted. For this reason, a tank that ensures the strength of each dome-shaped portion and that reduces the usage of fiber reinforced resin is desired.

(6) The disclosure provides a tank that ensures the strength of each dome-shaped portion and reduces the usage of fiber reinforced resin.

(7) An aspect of the disclosure relates to a tank. The tank includes a liner and a reinforcement layer. The liner has a body portion with a cylindrical shape, and a pair of dome-shaped portions respectively provided at both sides of the body portion in an axial direction of the body portion. The reinforcement layer covers the liner and is made of a fiber reinforced resin that is a resin-impregnated fiber bundle. The reinforcement layer includes a part reinforcement layer disposed from an apex side of each of the dome-shaped portions across a boundary between the dome-shaped portion and the body portion to part of the body portion, and an outer hoop layer disposed outside the part reinforcement layer and provided so as to press an end, adjacent to the body portion, of the part reinforcement layer.

(8) With the tank according to the aspect of the disclosure, since the part reinforcement layer is disposed from the apex side of each of the dome-shaped portions across the boundary between the dome-shaped portion and the body portion to part of the body portion, each of the dome-shaped portions is reinforced by the part reinforcement layer. For this reason, the amount of helical layer used to reinforce the dome-shaped portions is reduced, so an amount to form a helical layer on the body portion also reduces with formation of the helical layer. As a result, the strength of each of the

dome-shaped portions is ensured, and the usage of fiber reinforced resin is reduced as compared to an existing art.
(9) In the tank according to the aspect of the disclosure, the end of the part reinforcement layer may be inclined from an outside of the tank toward an inside of the tank such that a thickness of the end reduces.
(10) With this configuration, a step is less likely to be formed at the end of the part reinforcement layer, so the influence of the step on the outer hoop layer is reduced. Therefore, a gap between the part reinforcement layer and the outer hoop layer disposed outside the part reinforcement layer is reduced to enhance adhesion between the part reinforcement layer and the outer hoop layer, so interlayer peeling is suppressed. As a result, the end of the part reinforcement layer is stably pressed by the outer hoop layer, and the part reinforcement layer is reliably fixed.
(11) In the tank according to the aspect of the disclosure, the reinforcement layer may further include an inner hoop layer disposed inside the part reinforcement layer so as to sandwich the end of the part reinforcement layer with the outer hoop layer. With this configuration, the end of the part reinforcement layer is further stabilized by sandwiching the end of the part reinforcement layer with the outer hoop layer and the inner hoop layer.
(12) In the tank according to the aspect of the disclosure, the inner hoop layer may be made up of split elements having a cylindrical shape and prepared in advance. With this configuration, the inner hoop layer is easily manufactured.
(13) In the tank according to the aspect of the disclosure, both ends of the inner hoop layer in the axial direction each may be an inclined end inclined from an outside of the tank toward an inside of the tank such that a thickness of the inclined end reduces, and the end of the part reinforcement layer may be disposed adjacent to a center of the body portion with respect to the inclined end of the inner hoop layer.
(14) Since both ends of the body portion in the axial direction in the inner hoop layer are inclined ends in this way, a step is less likely to be formed at each inclined end, with the result that the influence of the step on the part reinforcement layer is reduced. Therefore, a gap between the inner hoop layer and the part reinforcement layer disposed outside the inner hoop layer is reduced to enhance adhesion between the inner hoop layer and the part reinforcement layer, so interlayer peeling is suppressed. Since the end of the part reinforcement layer is disposed adjacent to the center of the body portion with respect to the inclined end of the inner hoop layer, that is, the end of the part reinforcement layer is disposed on the body portion that is relatively flat, so pressing of the outer hoop layer against the end of the part reinforcement layer is stabilized.
(15) According to the aspect of the disclosure, it is possible to provide a tank that ensures the strength of each dome-shaped portion and that reduces the usage of fiber reinforced resin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Features, advantages, and technical and industrial significance of exemplary embodiments of the disclosure will be described below with reference to the accompanying drawings, in which like signs denote like elements, and wherein:
(2) FIG. **1** is a sectional view that shows a tank according to a first embodiment;
(3) FIG. **2** is a partially enlarged sectional view that shows the tank according to the first embodiment;
(4) FIG. **3** is a sectional view that shows a tank according to a second embodiment; and
(5) FIG. **4** is a partially enlarged sectional view that shows the tank according to the second embodiment.
DETAILED DESCRIPTION OF EMBODIMENTS

(6) Hereinafter, embodiments of a tank according to the disclosure will be described with reference to the accompanying drawings. Like reference numerals denote the same elements in the description of the drawings, and the description thereof will not be repeated. In the following description, an example in which the tank is mounted on a fuel cell electric vehicle and is filled with high-pressure hydrogen gas inside will be described. A gas allowed to be filled into the tank is not limited to hydrogen gas. Examples of the gas may include a compressed gas, such as compressed natural gas (CNG), and various liquefied gases, such as liquefied natural gas (LNG) and liquefied petroleum gas (LPG).

First Embodiment

(7) FIG. **1** is a sectional view that shows a tank according to a first embodiment. FIG. **2** is a partially enlarged sectional view that shows the tank according to the first embodiment. As shown in FIG. **1**, the tank **1** according to the present embodiment is a high-pressure gas storage container having a substantially cylindrical shape with both ends rounded in a dome shape. The tank **1** includes a liner **10**, a reinforcement layer **20**, an end fitting **30**, and a valve **40**. The liner **10** has gas barrier properties. The reinforcement layer is formed so as to cover the outer periphery of the liner **10**. The end fitting **30** is fitted to one end of the tank **1**. The valve **40** closes the end fitting **30**.

(8) The liner **10** is a hollow container having a storage space **2** for storing high-pressure hydrogen. The liner **10** is made of a resin material having gas barrier properties against hydrogen gas. The liner **10** is made up of a body portion **11** having a cylindrical shape, and a pair of dome-shaped portions (a first dome-shaped portion **12** and a second dome-shaped portion **13**) respectively provided at both ends of the body portion **11** in an axial direction (that is, an axis L direction of the tank **1**). The body portion **11** extends in the axis L direction of the tank **1** by a predetermined length. For example, the first dome-shaped portion **12** is disposed on the right side of the body portion **11**, and the second dome-shaped portion **13** is disposed on the left side of the body portion **11**. Each of the first dome-shaped portion **12** and the second dome-shaped portion **13** has a hemispherical shape such that the diameter reduces with distance from the body portion **11**. In the present embodiment, a boundary between first dome-shaped portion **12** having a hemispherical shape and the body portion **11** having a cylindrical shape is indicated by reference numeral **14**, and a boundary between the second dome-shaped portion **13** having a hemispherical shape and the body portion **11** having a cylindrical shape is indicated by reference numeral **15**.

(9) An opening is provided at the apex of one (in the present embodiment, the first dome-shaped portion **12**) of the dome-shaped portions, and the end fitting **30** integrally molded with the liner **10** is inserted through the opening. On the other hand, the second dome-shaped portion **13** has no opening. Alternatively, the second dome-shaped portion **13** may have an opening through which an end fitting **30** is inserted, as in the case of the first dome-shaped portion **12**.

(10) The thus configured liner **10** may be formed by forming a body split element, a first dome-shaped portion split element, and a second dome-shaped portion split element through injection molding, blow molding, or the like by using, for example, a resin material, such as polyethylene and nylon, and coupling these split elements, or may be integrally formed by rotational and blow molding by using a resin material, such as polyethylene and nylon.

(11) The reinforcement layer **20** has the function to reinforce the liner **10** and improve mechanical strength, such as stiffness and pressure resistance, of the tank **1**. The reinforcement layer **20** has a plurality of layers made of a fiber reinforced resin. The fiber reinforced resin is, for example, formed by impregnating a fiber bundle with a thermosetting resin or a thermoplastic resin. The fiber bundle is a bundle of fiber with a diameter of about several micrometers. Examples of the fiber may include carbon fiber, glass fiber, aramid fiber, alumina fiber, boron fiber, steel fiber, PBO fiber, natural fiber, and reinforced fiber, such as high-strength polyethylene fiber. Particularly, from the viewpoint of light weight, mechanical strength, and the like, carbon fiber is preferably used.

(12) Examples of the thermosetting resin include epoxy resin, modified epoxy resin as typified by vinyl ester resin, phenolic resin, melamine resin, urea resin, unsaturated polyester resin, alkyd

resin, polyurethane resin, and thermosetting polyimide resin. Examples of the thermoplastic resin include polyether ether ketone, polyphenylene sulfide, polyacrylic ester, polyimide, and polyamide.
(13) As shown in FIG. **1**, the reinforcement layer **20** includes an inner helical layer **21**, an inner hoop layer **22**, a first part reinforcement layer **24**, a second part reinforcement layer **25**, an outer hoop layer **26**, and an outer helical layer **27**. The inner helical layer **21** covers the whole of the liner **10**. The inner hoop layer **22** is disposed outside the inner helical layer **21** at a location corresponding to the body portion **11** of the liner **10**. The first part reinforcement layer **24** is disposed outside the inner helical layer **21** at a location corresponding to the first dome-shaped portion **12** of the liner **10**. The second part reinforcement layer **25** is disposed outside the inner helical layer **21** at a location corresponding to the second dome-shaped portion **13** of the liner **10**. The outer hoop layer **26** is formed so as to press an end **241** of the first part reinforcement layer **24** and an end **251** of the second part reinforcement layer **25**. The outer helical layer **27** is disposed outside the outer hoop layer **26** and formed so as to cover the whole of the liner **10**.
(14) The inner helical layer **21** is formed by helical winding of a resin-impregnated fiber bundle around the outer peripheries of the body portion **11**, first dome-shaped portion **12**, and second dome-shaped portion **13**.
(15) The inner hoop layer **22** is formed by hoop winding of a resin-impregnated fiber bundle at a location corresponding to the body portion **11** on the outer periphery of the inner helical layer **21**. As shown in FIG. **1** and FIG. **2**, in the axial direction (that is, the axis L direction of the tank **1**) of the body portion **11**, one end (for example, an end closer to the first dome-shaped portion **12**) of the inner hoop layer **22** is an inclined end **221** that is inclined from the outside of the tank **1** toward the inside such that the thickness reduces. The other end (for example, an end close to the second dome-shaped portion **13**) of the inner hoop layer **22** is an inclined end **222** that is inclined from the outside of the tank **1** toward the inside such that the thickness reduces.
(16) A method of forming the inclined ends **221**, **222** of the inner hoop layer **22** includes, for example, forming a first-layer hoop layer by hoop winding of a resin-impregnated fiber bundle from one end toward the other end in the axial direction of the body portion **11**, forming a second-layer hoop layer by hoop winding of the fiber bundle from the other end toward one end while shifting the fiber bundle such that a start point of the second-layer hoop layer is disposed adjacent to the center of the body portion **11** with respect to an end point of the first-layer hoop layer and shifting the fiber bundle such that an end point of the second-layer hoop layer is disposed adjacent to the center of the body portion **11** with respect to the start point of the first-layer hoop layer. After that, when third-layer, fourth-layer . . . Nth-layer hoop layers are sequentially shifted and laminated in the same manner, the inner hoop layer **22** having the inclined ends **221**, **222** is formed.
(17) The first part reinforcement layer **24** is a reinforcement layer mainly for the first dome-shaped portion **12**, and the second part reinforcement layer **25** is a reinforcement layer mainly for the second dome-shaped portion **13**. The two part reinforcement layers each have a dome shape and differ from each other in whether the layer has an opening through which the end fitting **30** is inserted. In other words, the first part reinforcement layer **24** has the opening through which the end fitting **30** is inserted, and the second part reinforcement layer **25** does not have an opening. Hereinafter, an example of the first part reinforcement layer **24** will be described, and the description of the second part reinforcement layer **25** is simplified.
(18) As shown in FIG. **1** and FIG. **2**, the first part reinforcement layer **24** is disposed from an apex side of the first dome-shaped portion **12** across the boundary **14** between the first dome-shaped portion **12** and the body portion **11** to part of the body portion **11**. Here, part of the body portion **11** is, for example, in the range of 100 mm to 200 mm from the boundary **14**. The end **241**, adjacent to the body portion **11**, of the first part reinforcement layer **24** is inclined from the outside of the tank **1** toward the inside such that the thickness reduces. The end **241** is disposed adjacent to the center of the body portion **11** with respect to the inclined end **221** of the inner hoop layer **22**.
(19) The first part reinforcement layer **24** is formed by placing a resin-impregnated fiber bundle on

part of the outer peripheries of the inner helical layer **21** and the inner hoop layer **22** in multiple layers with, for example, an automated fiber placement (AFP) method. The first part reinforcement layer **24** is formed by, for example, sticking prepreg made of carbon fiber reinforced plastics (CFRP) to part of the outer peripheries of the inner helical layer **21** and the inner hoop layer **22** by using an automated fiber placement machine.

(20) A method of forming the inclined end **241** of the first part reinforcement layer **24** includes, for example, forming a first layer and then forming a second layer on the first layer by sticking a resin-impregnated fiber bundle while shifting the fiber bundle such that a second-layer end point is disposed adjacent to the apex of the first dome-shaped portion **12** with respect to a first-layer end point. After that, when a third layer, a fourth layer . . . an Nth layer of fiber bundles are sequentially shifted and laminated in the same manner, the first part reinforcement layer **24** having the end **241** that is inclined is formed.

(21) The first part reinforcement layer **24** may be formed by using a mandrel having the same outside diameters as the inner helical layer **21** and the inner hoop layer **22**. In this case, the first part reinforcement layer **24** formed on the mandrel may be removed from the mandrel after being cured and then fitted to the inner helical layer **21** and the inner hoop layer **22**, or may be removed from the mandrel in an uncured state and then fitted to the inner helical layer **21** and the inner hoop layer **22**.

(22) However, a gap is hard to occur due to pressing with a roller in the automated fiber placement machine, so the first part reinforcement layer **24** is preferably formed by directly laminating a fiber bundle on the outer peripheries of the inner helical layer **21** and the inner hoop layer **22**. Since the occurrence of a gap is suppressed in this way, adhesion between the first part reinforcement layer **24** and each of the inner helical layer **21** and the inner hoop layer **22** improves, with the result that interlayer peeling is suppressed. The width of the resin-impregnated fiber bundle, used in the AFP method may be different from the width of a resin-impregnated fiber bundle, used in the FW method.

(23) As shown in FIG. **1**, the second part reinforcement layer **25** is disposed from an apex side of the second dome-shaped portion **13** across the boundary **15** between the second dome-shaped portion **13** and the body portion **11** to part of the body portion **11**. The end **251**, adjacent to the body portion **11**, of the second part reinforcement layer **25** is inclined from the outside of the tank **1** toward the inside such that the thickness reduces. The end **251** is disposed adjacent to the center of the body portion **11** with respect to the inclined end **222** of the inner hoop layer **22**.

(24) The outer hoop layer **26** is disposed outside the first part reinforcement layer **24** and the second part reinforcement layer **25** and is formed so as to press the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25**. Specifically, as shown in FIG. **1**, the outer hoop layer **26** is disposed at a location corresponding to the body portion **11** of the liner **10** in a state where the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** are pressed. The outer hoop layer **26** is formed over the entire length of the body portion **11**. The outer hoop layer **26** has a first portion **261**, a second portion **262**, and a third portion **263**. The first portion **261** covers part of the first part reinforcement layer **24** including the end **241**. The second portion **262** covers part of the second part reinforcement layer **25** including the end **251**. The third portion **263** covers the inner hoop layer **22** between the first part reinforcement layer **24** and the second part reinforcement layer **25**.

(25) An end of the first portion **261** is inclined from the outside of the tank **1** toward the inside such that the thickness reduces. Similarly, an end of the second portion **262** is inclined from the outside of the tank **1** toward the inside such that the thickness reduces. With this configuration, a step is less likely to be formed at any of these ends. Therefore, at the time of forming the outer helical layer **27** outside the outer hoop layer **26**, the influence of a step on the outer helical layer **27** is reduced, so interlayer peeling is reduced.

(26) The outer hoop layer **26** is, for example, formed by hoop winding of a resin-impregnated fiber

bundle on the outer peripheries of the first part reinforcement layer **24**, the second part reinforcement layer **25**, and the inner hoop layer **22** between the first part reinforcement layer **24** and the second part reinforcement layer **25** so as to press the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25**.
(27) The outer helical layer **27** is a layer located in the outermost side of the reinforcement layer **20** and is formed so as to cover the first part reinforcement layer **24**, the second part reinforcement layer **25**, and the outer hoop layer **26**. The outer helical layer **27** is, for example, formed by helical winding of a resin-impregnated fiber bundle on the outer peripheries of the first part reinforcement layer **24**, the second part reinforcement layer **25**, and the outer hoop layer **26**.
(28) On the other hand, the end fitting **30** is obtained by machining a metal material, such as a stainless steel and an aluminum alloy, into a predetermined shape. The end fitting **30** has an end fitting body portion **31** and a flange portion **32**. The end fitting body portion **31** has a substantially cylindrical shape and extends in the axis L direction of the tank **1**. The flange portion **32** is formed integrally with the end fitting body portion **31** and extends in a radial direction of the tank **1**. A communication hole **33** is provided in the end fitting body portion **31**. The communication hole **33** communicates with the storage space **2** of the tank **1**. An internal threaded portion for being screwed to the valve **40** is formed on an inner peripheral wall (that is, a portion forming the communication hole **33**) of the end fitting body portion **31**.
(29) The valve **40** is a member for filling and discharging hydrogen gas to and from the storage space **2** of the tank **1**. The valve **40** is made of a metal material, such as a stainless steel and an aluminum alloy. The valve **40** is inserted in the communication hole **33** so as to close the end fitting **30** and is screwed to the end fitting **30** with an external threaded portion formed on the outer peripheral wall of the valve **40**.
(30) The tank **1** according to the present embodiment includes the first part reinforcement layer **24** and the second part reinforcement layer **25**. The first part reinforcement layer **24** is disposed from the apex side of the first dome-shaped portion **12** across the boundary **14** between the first dome-shaped portion **12** and the body portion **11** to part of the body portion **11**. The second part reinforcement layer **25** is disposed from the apex side of the second dome-shaped portion **13** across the boundary **15** between the second dome-shaped portion **13** and the body portion **11** to part of the body portion **11**. Therefore, the first dome-shaped portion **12** is reinforced by the first part reinforcement layer **24**, and the second dome-shaped portion **13** is reinforced by the second part reinforcement layer **25**. For this reason, the amount of helical layer used to reinforce the dome-shaped portions is reduced, so an amount to form a helical layer on the body portion also reduces with formation of the helical layer. As a result, the strength of each of the first dome-shaped portion **12** and the second dome-shaped portion **13** is ensured, and the usage of fiber reinforced resin is reduced as compared to an existing art.
(31) Since the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** each are inclined from the outside of the tank **1** toward the inside such that the thickness reduces, a step is less likely to be formed at any of these ends. Thus, the influence of a step on the outer hoop layer **26** is reduced. Therefore, meandering of a fiber bundle is suppressed at the time of forming the outer hoop layer **26** outside the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25**. Hence, a gap between the first part reinforcement layer **24** and the outer hoop layer **26** disposed outside the first part reinforcement layer **24** and a gap between the second part reinforcement layer **25** and the outer hoop layer **26** disposed outside the second part reinforcement layer **25** are suppressed, so adhesion with the outer hoop layer **26** is enhanced, and interlayer peeling is suppressed. As a result, the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** each are stably pressed by the outer hoop layer **26**, so the first part reinforcement layer **24** and the second part reinforcement layer **25** are reliably fixed by the outer hoop layer **26**.
(32) The reinforcement layer **20** further includes the inner hoop layer **22** disposed inside the first

part reinforcement layer **24** and the second part reinforcement layer so as to sandwich the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** with the outer hoop layer **26**. With this configuration, by sandwiching the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** with the outer hoop layer **26** and the inner hoop layer **22**, the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** are reliably stabilized.

(33) Since both ends of the inner hoop layer **22** are the inclined ends **221**, **222**, a step is less likely to be formed at any of these inclined ends **221**, **222**, so the influence of a step on the first part reinforcement layer **24** and the second part reinforcement layer **25** is reduced. Therefore, a gap between the inner hoop layer **22** and the first part reinforcement layer **24** disposed outside the inner hoop layer **22** and a gap between the inner hoop layer **22** and the second part reinforcement layer **25** disposed outside the inner hoop layer **22** are reduced to enhance adhesion between the inner hoop layer **22** and the first part reinforcement layer **24** and adhesion between the inner hoop layer **22** and the second part reinforcement layer **25**, so interlayer peeling is suppressed.

(34) In addition, the end **241** of the first part reinforcement layer **24** is disposed adjacent to the center of the body portion **11** with respect to the inclined end **221** of the inner hoop layer **22**, and the end **251** of the second part reinforcement layer **25** is disposed adjacent to the center of the body portion **11** with respect to the inclined end **222** of the inner hoop layer **22**. With this configuration, the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** are disposed on the body portion **11** that is relatively flat, so the ends **241**, **251** are stably pressed by the outer hoop layer **26**.

Second Embodiment

(35) Hereinafter, a second embodiment of a tank will be described with reference to FIG. **3** and FIG. **4**. The tank **1**A according to the present embodiment differs from that of the first embodiment in that the inner helical layer **21** is not provided and an inner hoop layer **23** made up of split elements manufactured in advance is provided instead of the inner hoop layer **22**. Hereinafter, only the difference will be described.

(36) Specifically, a reinforcement layer **20**A includes the inner hoop layer **23**, the first part reinforcement layer **24**, the second part reinforcement layer **25**, the outer hoop layer **26**, and the outer helical layer **27**. The inner hoop layer **23** covers the body portion **11** of the liner **10**. The first part reinforcement layer **24** is disposed outside the inner hoop layer **23** and covers the first dome-shaped portion **12** and part of the inner hoop layer **23**. The second part reinforcement layer **25** is disposed outside the inner hoop layer **23** and covers the second dome-shaped portion **13** and part of the inner hoop layer **23**. The outer hoop layer **26** is formed so as to press the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25**. The outer helical layer **27** is disposed outside the outer hoop layer **26** and formed so as to cover the whole of the liner **10**.

(37) The inner hoop layer **23** is formed so as to directly cover the outer periphery of the body portion **11**. The inner hoop layer **23** is made up of split elements having a cylindrical shape and manufactured in advance and is fitted to the body portion **11** of the liner **10**. In the axial direction (that is, the axis L direction of the tank **1**A) of the body portion **11**, one end (for example, an end close to the first dome-shaped portion **12**) of the inner hoop layer **23** is an inclined end **231** that is inclined from the outside of the tank **1**A toward the inside such that the thickness reduces. The other end (for example, an end close to the second dome-shaped portion **13**) of the inner hoop layer **23** is an inclined end **232** that is inclined from the outside of the tank **1**A toward the inside such that the thickness reduces.

(38) The inner hoop layer **23** is, for example, formed by hoop winding of a resin-impregnated fiber bundle in multiple layers on the outer periphery of a mandrel having a cylindrical shape and then removing the layer from the mandrel after being cured. The mandrel is, for example, made of a metal and has the same outside diameter as the outside diameter of the body portion **11** of the liner

10.
(39) A method of forming the inclined ends **231**, **232** of the inner hoop layer **23** includes, for example, forming a first-layer hoop layer by hoop winding of a resin-impregnated fiber bundle from one end toward the other end in the axial direction of the mandrel, forming a second-layer hoop layer by hoop winding of the fiber bundle from the other end of the mandrel toward one end while shifting the fiber bundle such that a start point of the second-layer hoop layer is disposed adjacent to the center of the mandrel with respect to an end point of the first-layer hoop layer and shifting the fiber bundle such that an end point of the second-layer hoop layer is disposed adjacent to the center of the mandrel with respect to the start point of the first-layer hoop layer. After that, when third-layer, fourth-layer . . . Nth-layer hoop layers are sequentially shifted and laminated in the same manner, the inner hoop layer **23** having the inclined ends **231**, **232** is formed.
(40) The inner hoop layer **23** manufactured in this way is fitted to the body portion **11** of the liner **10** in a state where the liner **10** is inserted inside. To suppress misalignment, an adhesive or the like is preferably filled between the body portion **11** and the inner hoop layer **23**.
(41) In the present embodiment, the first part reinforcement layer **24** is formed so as to directly cover the first dome-shaped portion **12** of the liner **10**, and the second part reinforcement layer **25** is formed so as to directly cover the second dome-shaped portion **13** of the liner **10**. The end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** are sandwiched by the outer hoop layer **26** and the inner hoop layer **23**.
(42) With the tank **1**A according to the present embodiment, operation and advantageous effects similar to those of the first embodiment are obtained. In addition, since the inner hoop layer **23** is made up of split elements having a cylindrical shape and manufactured in advance, the inner hoop layer **23** is easily manufactured as compared to when an inner hoop layer is formed by directly winding a fiber bundle on the body portion **11**.
(43) Since both ends of the inner hoop layer **23** are the inclined ends **231**, **232**, a step is less likely to be formed at any of these inclined ends **231**, **232**, so the influence of a step on the first part reinforcement layer **24** and the second part reinforcement layer **25** is reduced. Therefore, a gap between the inner hoop layer **23** and the first part reinforcement layer **24** disposed outside the inner hoop layer **23** and a gap between the inner hoop layer **22** and the second part reinforcement layer **25** disposed outside the inner hoop layer **23** are reduced to enhance adhesion between the inner hoop layer **23** and the first part reinforcement layer **24** and adhesion between the inner hoop layer **23** and the second part reinforcement layer **25**, so interlayer peeling is suppressed.
(44) In addition, the end **241** of the first part reinforcement layer **24** is disposed adjacent to the center of the body portion **11** with respect to the inclined end **231** of the inner hoop layer **23**, and the end **251** of the second part reinforcement layer **25** is disposed adjacent to the center of the body portion **11** with respect to the inclined end **232** of the inner hoop layer **23**. In this way, when the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** are disposed on the body portion **11** that is relatively flat, the ends **241**, **251** are stably pressed by the outer hoop layer **26**.
(45) The embodiments of the disclosure have been described in detail; however, the disclosure is not limited to the above-described embodiments. Various design changes are applicable without departing from the spirit of the disclosure described in the appended claims.
(46) For example, in the above-described embodiments, an example in which the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25** are sandwiched by the outer hoop layer **26** and the inner hoop layer **22** or the inner hoop layer **23**, that is, an example in which the ends of the part reinforcement layers are sandwiched by two hoop layers have been described; however, the disclosure is not limited thereto. For example, a hoop layer may be disposed outside the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25**, and a helical layer may be disposed inside the end **241** of the first part reinforcement layer **24** and the end **251** of the second part reinforcement layer **25**.

Claims

1. A tank comprising: a liner having a body portion with a cylindrical shape, and a pair of dome-shaped portions respectively provided at both ends of the body portion in an axial direction of the body portion; and a reinforcement layer covering the liner and made of a fiber reinforced resin that is a resin-impregnated fiber bundle, wherein the reinforcement layer includes a part reinforcement layer disposed from an apex side of each of the dome-shaped portions across a boundary between the dome-shaped portion and the body portion to part of the body portion, and an outer hoop layer disposed outside the part reinforcement layer and provided so as to press an end, adjacent to the body portion, of the part reinforcement layer, wherein the reinforcement layer further includes an inner hoop layer disposed inside the part reinforcement layer so as to sandwich the end of the part reinforcement layer with the outer hoop layer, wherein an inclined end of the part reinforcement layer is made by forming a first layer and then forming a second layer on the first layer by sticking a resin-impregnated fiber bundle while shifting the fiber bundle such that an end point of the second layer is disposed adjacent to the apex side of the dome-shaped portion with respect to an end point of the first layer, and both ends of the inner hoop layer in the axial direction each are an inclined end inclined from an outside of the tank toward an inside of the tank such that a thickness of the inclined end reduces, the end of the part reinforcement layer extends axially toward a center of the body portion with respect to the inclined end of the inner hoop layer, and wherein the end of the part reinforcement layer is inclined from an outside of the tank toward an inside of the tank such that a thickness of the end reduces.